FAILURE MODE AND EFFECTS ANALYSIS (PROCESS FMEA)

METALMAN

Project name:-	952NX	Componenet Part No:- 92187158	PFMEA No.:-MAPL/PPW/CNH/PFMEA/050/92187158
Team Leader:	Mr. Satish Pandey	Component Description:-DOOR, ASSY, LH, 952NX	
Team Members:	Mr. Mr.SS/PRS/BY/DD/VNK/AV/AD/OM/SS/AJ	**Component Revision No.:- C**	PFMEA Last Review Date : 17-07-2025

Format No : F/NPD/07
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Revision No. : 00

PROTOTYPE | PRE-LAUNCH | PRODUCTION

No.	Process Function	Potential Failure Mode	Potential Effect(s) of Failure	Severity	CLASS	Potential Cause(s) of Failure	Occurrence	Current Controls Prevention	Current Controls Detection	Detection	RPN	Recommended Action	Responsibility & Completion Date	Action Taken	Revised Sev.	Revised Occ.	Revised Det.	Revised RPN
OP-10A						**SUB-ASSY -92187146-1 -Door WA,LH,952NX, Rev-C**												
10	Child Part Collection	Wrong child part used	M/C: Part may be not fitted SO: Operation. Effected CU: Fitment Issue EU: NIL			Wrong part taken		Part no. wise child part storage	Fixture Control		0							
20	Tack Welding	Visual Appearance	NO: Part visual appearance poor (4) SO: No Effect (1)			1. Weld feed rate high 2. Oil surface 3. Untrained Operator		1. Set up approval by attribute method	1. Operator Training for Welding		0 0 0							
		Weld Strength	NO: Part failure during testing, may be complete lot segregate (6) SO: No Effect (1) CE: Customer high dis-satisfied may be reject the			1. Process parameter variation 2. Oil surface 3. Less Weld Area 4. Fan Over Head		1. Set up approval by attribute method 4. Follow SOP FAN Caution while welding Format no F/NPD/11 & SOP no MAPL/FAN/01	1. SET UP INSPECTION OF PART 2. INPROCESS INSPECTION WITH BEND GAUGE		0 0 0 0							
30	Full Welding	Visual Appearance	NO: Part visual appearance poor (4) SO: No Effect (1)			1. Weld feed rate high 2. Oil surface 3. Untrained Operator		1. Set up approval by attribute method	1. Operator Training for Welding		0 0 0							
		Weld Strength	NO: Part failure during testing, may be complete lot segregate (6) SO: No Effect (1) CE: Customer high dis-satisfied may be reject the			1. Process parameter variation 2. Oil surface 3. Less Weld Area 4. Fan Over Head		1. Set up approval by attribute method 4. Follow SOP FAN Caution while welding Format no F/NPD/11 & SOP no MAPL/FAN/01	1. SET UP INSPECTION OF PART 2. INPROCESS INSPECTION WITH BEND GAUGE		0 0 0 0							
40	Finshing/Deburring	GRINDING MARKS ON THE SURFACE	M/C: NIL SO: PART MAY BE REJECTED CU: NIL EU: APPEARANCE NOT SMOOTH			1. LOW GRIT GRINDING WHEEL USED		1. USE HIGH GRIT GRINDING WHEEL	1. 100% VISUAL INSPECTION BY OPERATOR		0							
50	Move for Next Process																	
OP-10B						**SUB-ASSY -92187146-2 -Door WA,LH,952NX, Rev-C**												
10	Child Part Collection	Wrong child part used	M/C: Part may be not fitted SO: Operation. Effected CU: Fitment Issue EU: NIL			Wrong part taken		Part no. wise child part storage	Fixture Control		0							
20	Tack Welding	Visual Appearance	NO: Part visual appearance poor (4) SO: No Effect (1)			1. Weld feed rate high 2. Oil surface 3. Untrained Operator		1. Set up approval by attribute method	1. Operator Training for Welding		0 0 0							
		Weld Strength	NO: Part failure during testing, may be complete lot segregate (6) SO: No Effect (1) CE: Customer high dis-satisfied may be reject the			1. Process parameter variation 2. Oil surface 3. Less Weld Area 4. Fan Over Head		1. Set up approval by attribute method 4. Follow SOP FAN Caution while welding Format no F/NPD/11 & SOP no MAPL/FAN/01	1. SET UP INSPECTION OF PART 2. INPROCESS INSPECTION WITH BEND GAUGE		0 0 0 0							
30	Full Welding	Visual Appearance	NO: Part visual appearance poor (4) SO: No Effect (1)			1. Weld feed rate high 2. Oil surface 3. Untrained Operator		1. Set up approval by attribute method	1. Operator Training for Welding		0 0 0							
		Weld Strength	NO: Part failure during testing, may be complete lot segregate (6) SO: No Effect (1) CE: Customer high dis-satisfied may be reject the			1. Process parameter variation 2. Oil surface 3. Less Weld Area 4. Fan Over Head		1. Set up approval by attribute method 4. Follow SOP FAN Caution while welding Format no F/NPD/11 & SOP no MAPL/FAN/01	1. SET UP INSPECTION OF PART 2. INPROCESS INSPECTION WITH BEND GAUGE		0 0 0 0							
40	Finshing/Deburring	GRINDING MARKS ON THE SURFACE	M/C: NIL SO: PART MAY BE REJECTED CU: NIL EU: APPEARANCE NOT SMOOTH			1. LOW GRIT GRINDING WHEEL USED		1. USE HIGH GRIT GRINDING WHEEL	1. 100% VISUAL INSPECTION BY OPERATOR		0							
50	Move for Next Process																	
OP-20						**Full Welding -92187146-Door WA,LH,952NX, Rev-C**												
10	Child Part Collection	Wrong child part used	M/C: Part may be not fitted SO: Operation. Effected CU: Fitment Issue EU: NIL			Wrong part taken		Part no. wise child part storage	Fixture Control		0							
20	Full Welding	Visual Appearance	NO: Part visual appearance poor (4) SO: No Effect (1)			1. Weld feed rate high 2. Oil surface 3. Untrained Operator		1. Set up approval by attribute method	1. Operator Training for Welding		0 0 0							
		Weld Strength	NO: Part failure during testing, may be complete lot segregate (6) SO: No Effect (1) CE: Customer high dis-satisfied may be reject the			1. Process parameter variation 2. Oil surface 3. Less Weld Area 4. Fan Over Head		1. Set up approval by attribute method 4. Follow SOP FAN Caution while welding Format no F/NPD/11 & SOP no MAPL/FAN/01	1. SET UP INSPECTION OF PART 2. INPROCESS INSPECTION WITH BEND GAUGE		0 0 0 0							
30	Finshing/Deburring	GRINDING MARKS ON THE SURFACE	M/C: NIL SO: PART MAY BE REJECTED CU: NIL EU: APPEARANCE NOT SMOOTH			1. LOW GRIT GRINDING WHEEL USED		1. USE HIGH GRIT GRINDING WHEEL	1. 100% VISUAL INSPECTION BY OPERATOR		0							
40	Move for Next Process																	
OP-30						**Painting (DOOR, ASSY, LH, 952NX-92244728) (Power Tan -Std-86628042)**												
10	Transportation to pretreatment (Material through trolly)	Material Spillage, wrong placement	Material Loss and Damage			Material Overflow 1. Wrong stacking 2. Un Skilled Transport operator LR Crane		SOP 1. Stack Designed 2. 100% visual insp. by operator Periodic insp. by inspector	Area wise 5S audit & CLR check sheet As per production plan, and PO. Spec Sheet, random check		0 0							
20	Knock of Degrease	Total alkeli less than 70 ml	Surface not cleaned properly			Due to continueous production Bonderite AK - 5055 is less		Inspection of the Titration once in every shift before start of the process	Inspection of the Titration once in every shift before start of the process		0							
		Total alkeli less than 90 ml	Water Brake			More Bonderite Ak - 5055 is mixed by operator		1.Inspection of the Titration once in every shift before start of the process. 2.training given to operator for material mixup	Inspection of the Titration once in every shift before start of the process		0							
		Temparature less than 50°C	Surface not cleaned properly			1. Addition of the AKVN water. 2. Heater not working.		1.Daily checking of the temperature before start of the process. 2.Controller stop the process automatically if temperatre getting down.	Visually on pannel display		0							
		Pressure of water less than 0.8 kg/Cm2	Part cleaning not done properly			Filter bag choke due to dust		1. Visuall inspection by the operator in pressure gauge beore start of the process and 2 times visually in every shift	1. Visuall inspection by the operator in pressure gauge beore start of the process and 2 times visually in every shift		0							
		Oil content more than 4gm/Ltr.	Part surface should be oily			1. Oil separator off 2. Oil separator not working properly		Using oil seperator to remove oil from water	Tank water inspection in third party lab once in every month		0							
30	Degrease	Total alkeli less than 70 ml	Surface not cleaned properly			Due to continueous production Bonderite AK - 5055 is less		Inspection of the Titration once in every shift before start of the process	Inspection of the Titration once in every shift before start of the process		0							
		Total alkeli less than 90 ml	Water Brake			More Bonderite Ak - 5055 is mixed by operator		1.Inspection of the Titration once in every shift before start of the process. 2.training given to operator for material mixup	Inspection of the Titration once in every shift before start of the process		0							
		Temparature less than 50°C	Surface not cleaned properly			1. Addition of the AKVN water. 2. Heater not working.		1.Daily checking of the temperature before start of the process. 2.Controller stop the process automatically if temperatre getting down.	Visually on pannel display		0							
		Pressure of water less than 0.8 kg/Cm2	Part cleaning not done properly			Filter bag choke due to dust		1. Visuall inspection by the operator in pressure gauge beore start of the process and 2 times visually in every shift	1. Visuall inspection by the operator in pressure gauge beore start of the process and 2 times visually in every shift		0							
		Oil content more than 4gm/Ltr.	Part surface should be oily			1. Oil separator off 2. Oil separator not working properly		Using oil seperator to remove oil from water	Tank water inspection in third party lab once in every month		0							
40	Water Rinse 1	Removal of alkali content	Customer dissatisfaction due to appearance not ok			Improper cleaning of material		1. Inspection of the Alkali once in every shift	1. Inspection of the Alkali once in every shift		0							
50	Water Rinse 2	Removal of alkali content	Customer dissatisfaction due to appearance not ok			Improper cleaning of material		1. Inspection of the Alkali once in every shift	1. Inspection of the Alkali once in every shift		0							
60	Surface Activation	Improper Surface activation	Customer dissatisfaction due to appearance not ok			1) Improper cleaning of material 2) Chemical concentration Not in Range		1. volumetric analysis, 2. pH	Check all the parameters of bath once in a shift		0							
70	Phosphate	Uneven phosphate layer on job	Customer dissatisfaction due to appearance not ok			1) Improper cleaning of material 2) Improper parameters of the phospate process		1. Check all the parameters of bath once in a shift 2. Temp. of bath.	Check all the parameters of bath once in a shift		0							
80	Water Rinse 03	Traces of acid	Customer dissatisfaction due to appearance not ok			Improper cleaning of material		1. volumetric analysis	Check all the parameters of bath once in a shift		0							
90	Water Rinse 04	Traces of acid	Customer dissatisfaction due to appearance not ok			Improper cleaning of material		1. volumetric analysis	Check all the parameters of bath once in a shift		0							
100	PTRCDM	Cleaning of component not ok	Customer dissatisfaction due to appearance not ok			1) Improper cleaning of material. 2) Chemical concentration Not in Range		Check all the parameters of bath once in a shift	Check all the parameters of bath once in a shift		0							
110	CED Coating	Dust, rupture, craters, lumps,glassbid uncover,scratches etc.	Customer dissatisfaction due to appearance not ok			Improper CED coating		1. Manitain level of the pH & conductivity of bath sample, NVM, voltage, current of bath as per requirement.	Check all the parameters of bath once in a day		0							
120	Ultra Filtration	Removal of Excess paint not done properly	Customer dissatisfaction due to appearance not ok			Improper Removal of Excess paint		1. Inspection and maintain pH of bath.	Check all the parameters of bath once in a day		0							
130	EDRCDM	Removal of Excess paint not done	Customer dissatisfaction due to appearance not ok			Removal of Excess paint		1. Maintain conductivity and ph of Bath	Check all the parameters of bath once in a day		0							
140	Unloading	Dust, rupture,craters, lumps,glassbid uncover,scratches etc.	Customer dissatisfaction due to appearance not ok			Ununiform Coating of component		Defect free component	100 % visual inspection		0							
150	Baking	Temperature is less/more than required	1.Poor Adhesion of paint 2.bubbles formation corrosion may occur. 3.Aesthetical Issues part may reject 4.Aesthetical Issue Corrosion will occur			1. Correct timing is not given for the process 2.Parameter of Certain proces is not correctly followed 3.unwanted chemical reactions may occur further		Check control panel setting and conveyor speed	Auto cut off system and temp recording		0							

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
160	Putty Application & Finishing	Surface not cleaned properly	Asthetic issue			Unskilled operator		Skill Matrix.	Inprocess Inspection		0							
170	Cleaning	Air pressure low in air cleaning	Customer dissatisfaction due to appearance not ok			1. Air leakage 2. Wrong air pressure set by regulator 3. Compressure not working		100% Door Assembly visual inspection as per before top coat check sheet	100% Door Assembly visual inspection as per before top coat check sheet		0							
		Tegorat cloth not cleaning the Door Assembly surface properly	Customer dissatisfaction due to appearance not ok			1. Wrong grade material used 2. Terogat cloth used in more than 2 Door Assembly s		100% Door Assembly visual inspection as per before top coat check sheet	100% Door Assembly visual inspection as per before top coat check sheet		0							
180	Inspection (Quality Gate No.-2 & 3)	Sealant missing	Leakage will be apear in next process or at customer end			1. New operator 2. SOP not displayed at shop floor 2. inspection not done after selant pasting		1.Sealant SOP in shop floor 2.Skill matrix	100% inspection of the all Door Assembly s as per selant check sheet		0							
		Finishing not ok	Paint peel off			Cleaning not done or unproper cleaning		1. Air cleaning not done 2. Terogat cleaning not done	100% inspection of the all Door Assembly s as per before top coat inspection check sheet		0							
190	Final Paint	LOW/MORE DFT	M/C: NIL SO: NIL CU: Part may rejected EU: Durability effected			*COATING NOT DONE AS PER STANDARD * UNTRAINED PAINTER		*COATING TO BE DONE AS PER WI *ONLY TRAINED PAINTER WILL DO THE COATING	*PROCESS AUDIT AT SUPPLIER AT SUPPLIER END *DFT METER * SKILL MATRIX FOR PAINTER ADHERE AS PER PLAN		0							
		GLOSS LOW/ HIGH	M/C: NIL SO: NIL CU: Part may rejected EU: Aesthetically rejected			*COATING NOT DONE AS PER STANDARD * UNTRAINED PAINTER		*COATING TO BE DONE AS PER WI *ONLY TRAINED PAINTER WILL DO THE COATING	*PROCESS AUDIT AT SUPPLIER AT SUPPLIER END *DFT METER * SKILL MATRIX FOR PAINTER ADHERE AS PER PLAN		0							
		PAINT DEFECT (PIN HOLE, PAINT PEEL OFF, OVER FLOW, ORANGE PEEL, EXCESS PAINT DEPOSITION/SHADE VARIATION)	M/C: NIL SO: NIL CU: Part may rejected EU: Aesthetically rejected			*COATING NOT DONE AS PER STANDARD * UNTRAINED PAINTER		*COATING TO BE DONE AS PER WI *ONLY TRAINED PAINTER WILL DO THE COATING	*PROCESS AUDIT AT SUPPLIER AT SUPPLIER END *DFT METER * SKILL MATRIX FOR PAINTER ADHERE AS PER PLAN		0							
		DENT & SERATCH ON PART	M/C: NIL SO: NIL CU: Part may rejected EU: Aesthetically rejected			*PACKAGING STANDARAD NOT ADHERE *POOR HANDLING OF PARTS *UNSKILLED MATERIAL HANDLING PERSON		*PACKAGING STANDARAD SHOULD BE ADHERED	*PART MOVEMENT IN PROPER TROLLYS *ONLY TRAINED PERSON HAS LOADING AND UNLOADING THE MATERIAL *MATERIAL HANDLING DONE AS PER WI * SKILL MATRIX OF MATERIAL LOADING AND UNLOADING PERSON ADHERED AS PER PLAN *PROCESS AUDIT AT SUPPLIER AT SUPPLIER END		0							
200	Backing	Temperature is less/more than required	No Effect			correct timing is not given for the process		Controlled as per SOP, Correct time to time maintainence Audits	None		0							
						Parameter of Certain proces is not correctly followed unwanted chemical reactions may occur further		Controlled as per SOP, Poka Yoke correctly installed	None		0							
						Un Skilled Transport operator operator neglegence		100% visual insp. by operator Periodic insp. by inspector, MSA Plan Training plan, Skill matrix	None		0							
						Wrong lot transportation Door Assembly must correctly aligned for correct accumulation of Hot water spray over the Door Assembly surface		Controlled as per SOP, Correct time to time maintainence Audits	None		0							
						Un Skilled Transport operator operator neglegence		100% visual insp. by operator Periodic insp. by inspector, MSA Plan Training plan, Skill matrix	None		0							
210	Inspection (Quality Gate No.-4)	DFT Found less	Part will be rusty			unskilled operator		Skill Matrix	None		0							
		DFT Found more	Paint peel off will be came			Unskilled operator		Skill Matrix	None		0							
		Painting defects	Asthetic problems			1.Surface preparation not done properly, 2.Paint not done as per SOP		Ispection by operator before final paint	None		0							
220	Storage	* STORAGE ON WRONG LOCATION / MIX-UP	M/C: NIL SO: Part may reject CU: Part Failure EU: Field Failure			1.No proper place for storage of material by grade and thick		*Storage area is identified as per grade and thickness wise.	1. WI Displayed near Raw material Storage area and Incoming Inspector Gives colour code after Inspection		0							
						2.No identification on sheets		2.Operator will use the material as per the color coding / FIFO system.	None		0							
230	Move for Next Process																	
	OP-40					**FINAL ASSEMBLY (92187158-DOOR, ASSY, LH, 952NX) Rev-C**												
10	Child Part Collection	Wrong child part used	M/C: Part may be not fitted SO: Operation. Effected CU: Fitment Issue EU: NIL			Wrong part taken		Part no. wise child part storage	Fixture Control		0							
20	RUBBER TRACK	Parts found loose	M/C:-NIL SO :-NIL CU:-Customer Complaint EU-Failure			Beading not proper fit		Air hammer available for beading fit	Inspection after semi dress		0							
		Part over tight	M/C:-NIL SO :-NIL CU:-Customer Complaint EU-Failure			Beading location shift		Beading Start point define in the part	1. SET UP INSPECTION OF PART 2. INPROCESS INSPECTION WITH BEND GAUGE		0							
30	BEADING-ASSEMBLY	Parts found loose	M/C:-NIL SO :-NIL CU:-Customer Complaint EU-Failure			Beading not proper fit		Air hammer available for beading fit	Inspection after semi dress		0							
		Part over tight	M/C:-NIL SO :-NIL CU:-Customer Complaint EU-Failure			Beading location shift		Beading Start point define in the part	1. SET UP INSPECTION OF PART 2. INPROCESS INSPECTION WITH BEND GAUGE		0							
40	Move for Next Process																	
	OP-50					**FINAL INSPECTION (92187158-DOOR, ASSY, LH, 952NX) Rev-C**												
10	Inspection (Quality Gate no.-5)	Sealant missing	Leakage will be apear in next process or at customer end			1. New operator 2. SOP not displayed at shop floor 2. inspection not done after selant pasting		1.Sealant SOP in shop floor 2.Skill matrix	100% inspection of the all Door Assembly s as per selant check sheet		0							
		Finishing not ok	Paint peel off			Cleaning not done or unproper cleaning		1. Air cleaning not done 2. Terogat cleaning not done	100% inspection of the all Door Assembly s as per before top coat inspection check sheet		0							
20	Finishing/Cleaning	Door Assembly not cleaned	M/C:-NIL SO :-NIL CU:-Customer complaints EU-NIL			Cleaning not done or improper cleaning by operator		100% inspection of the Door Assembly at PDI stage by QC operator as per PDI check sheet	100% inspection of the Door Assembly at PDI stage by QC operator as per PDI check sheet		0							
30	Pre-Dispatch Inspection	WRONG INSTRUMENT USED	M/C:-NIL SO :-NIL CU:-Fitment Issue EU-Durability Effected			LACK OF KNOWLEDGE		PROVIDE PDI TRAINING TO Q.C. ENGINEER	FINAL INSPECTION & FIREWALL INSPECTION		0							
		NG PART PASSED	M/C:-NIL SO :-Dim. Effected CU:-Fitment Issue EU-NIL			LACK OF KNOWLEDGE		1. ADD FITMENT DIM IN PQS BEFORE SIGN OFF 2. CALIBRATE & REVALIDATE GAUGE & INSTRUMENT AS PER PLAN	FINAL INSPECTION / FIREWALL INSPECTION BEFORE LOT DISPATCH		0							
40	Move to Dispatch	PART DAMAGED	M/C:-NIL SO :-Dim. Effected CU:-Fitment Issue EU-NIL			1. IMPROPER LOADING / UNLOADING 2. NOT PROPER STACKING OF PARTS		1. FOLLOW PACKING STD 2. LOADING/UNLOADING IS DONE BY SKILLED PERSON	IT SHOULD BE ENSURE THAT PROPER STACKING OF PART IS DONE BEFORE DISPATCH		0							
		QTY VARIATION	M/C:-NIL SO :-NIL CU:-CUSTOMER DISSATISFACTION EU-NIL			COUNTING OF PART IS NOT DONE WITH CARE		FOLLOW SCHEDULE / SUPPLY	VERIFY QTY OF PART AS PER SCHEDULE		0							

ABBREVIATION- M/C:- MACHINE SO:-SECOND OPERATION CU:- CUSTOMER USE EU:- EXTERNAL USE

Prepared By : Mr.Amit Vishwakarma **Approved By- Mr.Prashant Raj Srivastava**

DATE	REV NO.	CHANGES	Prepared By	Approved By
17-Jul-25	A	Initial for revision no. "C"	Mr.Amit Vishwakarma	Mr. Prashant Raj Srivastava

Note

PFMEA UPDATE Frequency : During customer complaint / Any process change / Yearly review

ACTION PLAN RULE : Action shall we taken if any RPN value crossed over 100 otherwise no need to take any action

FAILURE MODE AND EFFECTS ANALYSIS (PROCESS FMEA)

METALMAN

Format No : F/NPD/07
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Project name:-	952NX	Componenet Part No:- 92187158	PFMEA No.:-MAPL/PPW/CNH/PFMEA/ 050/92187158
Team Leader:	Mr. Satish Pandey	Component Description:-DOOR, ASSY, LH, 952NX	Revised on : 00
Team Members:	Mr. Mr.SS/PRS/BY/DD/VNK/AV/AD/OM/SS/AJ	**Component Revision No.:- C**	Revision No. : 00
			PFMEA Last Review Date : 17-07-2025

PROTOTYPE				PRE-LAUNCH								PRODUCTION							
No.	**Process Function**	**Potential Failure Mode**	**Potential Effect(s) of Failure**	**Severit**	**CLASS**	**Potential Cause(s) of Failure**	**ccurren**	**Current Controls Prevention**	**Current Controls Detection**	**tect**	**RPN**	**Recommended Action**	**Responsibility & Completion Date**	**Action Taken**	**Revised Sev.**	**Occ.**	**Det.**	**RPN**
Sheet Metal Parts PFMEA																			
10	RM Procurement	*MATERIAL FROM APPROVED SOURCE	M/C:- NIL SO :- Operation. Effected CU:- Part Failure EU- M/C Breakdown			Material from local market		Material from Approved source	Approved Source/ supplier list										
20	INCOMING INSPECTION	* THICKNESS OVER SIZE	M/C:- NIL SO :- Bending Dim. Effected CU:- Fitment Issue EU- NIL			1. Wrong PO / Incomplete Details in PO Issued to supplier.		1. Verification by Sign. Authority of PO with Purchase indent and before issue to supplier.	1. MATERIAL CROSS VERIFY BY QA INSPECTOR BEFORE ENTERING THE PLANT AGAINST MTC & INVOICE.										
						2. Wrong PO / Incomplete Details in PO Issued to supplier.		2. Check at receiving stage	2. Mill TC should be verified by field inspector.										
						3. No verification of TC by QA inspector.		3. Third party material test Plan to be adhered.	3. Verification of Mill TC material by MRN and third party test Report.										
						4. NON CALIBRATED INSTRUMENT USE FOR INSPECTION		4. Instrument calibration plan to be adhered.	4. Calibrated instrument should be used to check the material.										
		* THICKNESS UNDER SIZE	M/C:- NIL SO :- Bending Dim. Effected CU:- Fitment Issue EU- NIL			1. Wrong PO / Incomplete Details in PO Issued to supplier.		1. Verification by Sign. Authority of PO with Purchase indent and before issue to supplier.	1. MATERIAL CROSS VERIFY BY QA INSPECTOR BEFORE ENTERING THE PLANT AGAINST MTC & INVOICE.										
						2. Wrong TC Provided by Supplier.		2. Check at receiving stage	2. Mill TC should be verified by field inspector.										
						3. No verification of TC by QA inspector.		3. Third party material test Plan to be adhered.	3. Verification of Mill TC material by MRN and third party test Report.										
						4. NON CALIBRATED INSTRUMENT USE FOR INSPECTION		4. Instrument calibration plan to be adhered.	4. Calibrated instrument should be used to check the material.										
		* PLATE RUSTY	M/C:- NIL SO :- Poor surface Treatment CU:- Fail in test EU- Durability Effected			1. Material Stored in open condition.		1. Material Stored in covered area. Awareness about it to Quality inspector.	1. Incoming Inspection Report. Before next process operator will ensure that Plate is free from RUST.										
						2. FIFO Not maintained		2. FIFO system applied	2. Color coding to be done on the steel plates. Green, Yellow.										
		* SHEET BEND	M/C:- NIL SO :- Bending Dim Effected CU:- Fitment Issue EU- Durability Effected			1. Plate may be hit / damage during unloading.		1. PROPER LOADING & UNLOADING BY HOST (CRANE)	1. Incoming Inspection Report										
						2. Plate may not be stored properly		2. STORAGE LOCATION DEFINED FOR RAW MATERIAL	2. Before next process start operator should ensure that Plate is not bend / Inprocess inspection Report.										
		* MATERIAL GRADE WRONG	M/C:- NIL SO :- Bending Effected CU:- Part Crack EU- Durability Effected			1. Wrong PO / Incomplete Details in PO Issued to supplier.		1. Verification by Sign. Authority of PO with Purchase indent and before issue to supplier.	1. Incoming Inspection Report										
						2. Wrong TC Provided by Supplier.		2. Verification of TC by Field inspector.											
						3. Chemical and Physical properties not matched with std		3. Chemical and Physical properties be checked & matched with std.	3. Third party test to be done as per plan.										
30	Storage	* STORAGE ON WRONG LOCATION / MIX-UP	M/C:- NIL SO :- Part may reject CU:- Part Failure EU- Field Failure			1. No proper place for storage of material by grade and thickness.		1. Storage area is Identified as per grade and thickness wise.	1. WI Displayed near Raw material Storage area and Incoming Inspector Gives colour code after Inspection.										
						2. No identification on sheets			2. Operator will use the material as per the color coding / FIFO system.										
40	Laser Cutting	TAPPER CUTTING	M/C:- NIL SO :- Part may reject, Excess rework CU:- Fitment issue EU- NIL			1. M/c parameter may not be set as per std.		1. M/c process parameter are to be set as per std. before process starts	1. Inprocess Inspection Report with Full Size Template. 2. Daily Process parameter check sheet.										
						2. Bed may not be clean.		2. Bed cleaning to be done as per decided frequency.	2. Bed cleaning date to be mention on bed.										
		* PROFILE OVER SIZE	M/C:- NIL SO :- Effect on bend operating CU:- Fitment issue EU- Durability Effected			1. Part Nesting is not ok.		1. Verification of nesting of machine before process starts.	1. 1st set up Inspection of part with Full Size Template.										
						2. Process parameter of m/c is not ok as per Std.		2. Reset process parameter of plasma m/c	2. Inprocess Inspection Report with Full Size Template.										
		* PROFILE UNDER SIZE	M/C:- NIL SO :- Part reject CU:- Fitment issue EU- NIL			1. Part Nesting is not ok		1. Verification of nesting of machine before process starts	1. 1st set up Inspection of part with Full Size Template.										
						2. Process parameter of m/c is not ok as per Std.		2. Reset process parameter of plasma m/c	2. Inprocess Inspection Report with Full Size Template.										
		* UNEVEN CUTTING / SLAG	M/C:- NIL SO :- Effect on Bend operation CU:- Fitment issue EU- NIL			Process parameter of m/c is not ok as per Std.		Reset process parameter of plasma m/c	1. 1st set up Inspection of part 2. Inprocess Inspection Report with Full Size Template										
50	Oxy Cutting	TAPPER CUTTING	M/C:- NIL SO :- Part may reject, Excess rework CU:- Fitment issue EU- NIL			1. M/c parameter may not be set as per std.		1. M/c process parameter are to be set as per std. before process starts	1. Inprocess Inspection Report with Full Size Template. 2. Daily Process parameter check sheet.										
						2. Bed may not be clean.		2. Bed cleaning to be done as per decided frequency.	2. Bed cleaning date to be mention on bed.										
		* PROFILE OVER SIZE	M/C:- NIL SO :- Effect on bend operating CU:- Fitment issue EU- Durability Effected			1. Part Nesting is not ok.		1. Verification of nesting of machine before process starts.	1. 1st set up Inspection of part with Full Size Template.										
						2. Process parameter of m/c is not ok as per Std.		2. Reset process parameter of plasma m/c	2. Inprocess Inspection Report with Full Size Template.										
		* PROFILE UNDER SIZE	M/C:- NIL SO :- Part reject CU:- Fitment issue EU- NIL			1. Part Nesting is not ok		1. Verification of nesting of machine before process starts	1. 1st set up Inspection of part with Full Size Template.										
						2. Process parameter of m/c is not ok as per Std.		2. Reset process parameter of plasma m/c	2. Inprocess Inspection Report with Full Size Template.										
		* UNEVEN CUTTING / SLAG	M/C:- NIL SO :- Effect on Bend operation CU:- Fitment issue EU- NIL			Process parameter of m/c is not ok as per Std.		Reset process parameter of plasma m/c	1. 1st set up Inspection of part 2. Inprocess Inspection Report with Full Size Template										
60	Drilling	DRILL UNDER OR OVER SIZE	M/C:- NIL SO :- Part reject CU:- Fitment issue EU- NIL			WRONG SIZE DRILL SELECTION		DRILL SIZE TO BE CHECK BEFORE USE	DIMENSIONS ARE CHECKED BY OPERATOR AFTER COMPLETING OPERATION.										
70	Tapping	TAPPING THREAD DEPTH MAY NOT BE CORRECT	M/C:- NIL SO :- Part reject CU:- Fitment issue EU- NIL			ALL 03 TAP NOT RUN IN TAPPING HOLE.		TAPPING HOLE TO BE CHECK AFTER OPERATION COMPLETE. THROUGH GO NO GO GAUGE	TAPPING HOLE CHECKED BY GO NO GO GAUGE.										
80	Chamfering	Less Chamfer	M/C:- NIL SO :- Difficulty for part fitment CU:- NIL EU- Difficulty for part fitment			1. Wrong drill tool selected 2. Untrained operator		1. SOP 2. Skill Matrix	100% VISUAL INSPECTION BY OPERATOR										
		Chamfer Over Size	M/C:- NIL SO :- Difficulty for part fitment CU:- NIL EU- Difficulty for part fitment			1. Wrong drill tool selected 2. Untrained operator		1. SOP 2. Skill Matrix	100% VISUAL INSPECTION BY OPERATOR										
90	Straightening	PART DAMAGE	M/C:- ROLLER CAN BE DAMAGE SO :- Part may reject CU:- Fitment Issue EU- Durability			OTHER MATERIAL FROM PART LIKE CHIPS, CUTTING SLUG.		MATERIAL CLEAN PROPERLY	INSPECTION AS PER WI										
		BOW	M/C:- NIL SO :- Part may reject CU:- Fitment Issue EU- Durability			1. WRONG METHOD USE FOR STRAIGHTENING 2. OPERATOR NEGLIGENCY		ONLY TRAINED OPERATOR DONE THE PROCESS	INPROCESS INSPECTION. SKILL MATRIX FOR OPERATOR ADHERE AS PER PLAN										
100	Bending/ Forming	ANGLE NOT OK	M/C:- NIL SO :- Part may be reject CU:- Fitment issue at customer end EU- Durability Effected			1. LOW HYD PRESSURE.		1. APPROPRIATE PRESSURE APPLY 2. PRESSURE GAUGE SHOULD WORK PROPERLY.	1. PRESSURE SHOULD BE APPLIED AS PER WORK INSTRUCTION. 2. PRESSURE GAUGE SHOULD BE CALIBRATED AS PER PLAN.										
						2. HIGH HYDRAULIC PRESSURE.		1. APPROPRIATE PRESSURE APPLY 2. PRESSURE GAUGE SHOULD WORK PROPERLY.	1. PRESSURE SHOULD BE APPLIED AS PER WORK INSTRUCTION. 2. PRESSURE GAUGE SHOULD BE CALIBRATED AS PER PLAN.										
		BEND HEIGHT IS NOT OK	M/C:- NIL SO :- Part may be reject CU:- Fitment issue at customer end EU- Durability Effected			1. INAPPROPRIATE HYD PRESSURE.		APPROPRIATE PRESSURE APPLY ON THE PART AS PER WORK INSTRUCTION.	1. SET UP INSPECTION OF PART 2. INPROCESS INSPECTION WITH BEND GAUGE										
		TOOL MARK	M/C:- NIL SO :- NIL CU:- Aesthetically reject EU- NIL			1. DIE & PUNCH WEAR OUT 2. ROUGH SURFACE OF TOOL		1. PREVENTIVE MAINTENANCE OF TOOL AS PER PLAN.	1. 1ST SET UP INSPECTION OF PART 2. 100% VISUAL INSPECTION BY OPERATOR										
		* MATERIAL CRACK	M/C:- Operator may be injured SO :- Part may reject CU:- Part Failure EU- Field Failure			1. PROPERTIES OF PLATE MAY BE NOT AS PER STD. 2. PLATE MAY BE LAMINATED.		1. PLATE COMPOSITION VERIFY DURING INCOMING INSPECTION 2. 3RD PARTY PLATE TESTING AS PER PLAN											
110	Storage	* STORAGE ON WRONG LOCATION / MIX-UP	M/C:- NIL SO :- Part may reject CU:- Part Failure EU- Field Failure			1. No proper place for storage of material by grade and thickness.		1. Storage area is Identified as per grade and thickness wise.	1. WI Displayed near Raw material Storage area and Incoming Inspector Gives colour code after Inspection.										
						2. No identification on sheets			2. Operator will use the material as per the color coding / FIFO system.										

120	Move for Next Process																				
ABBREVIATION-	M/C:- MACHINE	SO:-SECOND OPERATION	CU:- CUSTOMER USE	EU:- EXTERNAL USE																	
Prepared By : Mr.Amit Vishwakarma									**Approved By- Mr.Prashant Raj Srivastava**												
DATE	**REV NO.**	**CHANGES**											**Prepared By**				**Approved By**				
17-Jul-25	**A**	**Initial for revision no. "C"**											**Mr.Amit Vishwakarma**				**Mr. Prashant Raj Srivastava**				
Note	PFMEA UPDATE Frequency : During customer complaint / Any process change / Yearly review																				
	ACTION PLAN RULE : Action shall we taken if any RPN value crossed over 100 otherwise no need to take any action																				

FAILURE MODE AND EFFECTS ANALYSIS (PROCESS FMEA)

METALMAN

Project name:-	952NX	Componenet Part No.:- 92187158	PFMEA No.:-MAPL/PP/WC/NH/PFMEA/ 050/92187158
Team Leader:	Mr. Satish Pandey	Component Description:-DOOR, ASSY, LH, 952NX	
Team Members:	Mr. Mr.SS/PRS/BY/DD/VNK/AV/AD/OM/SS/AJ	Component Revision No.:- C	PFMEA Last Review Date : 17-07-2025

Format No : F/NPD/07
Released on : 01/07/2022
Revised on : 00
Revision No. : 00

PROTOTYPE	PRE-LAUNCH	PRODUCTION

No.	Process Function	Potential Failure Mode	Potential Effect(s) of Failure	Severit	CLAS	Potential Cause(s) of Failure	curre	Current Controls Prevention	Current Controls Detection	ect	RPN	Recommended Action	Responsibility & Completion Date	Action Taken	Revised Sev.	Revised Occ.	Revised Det.	Revised RPN
BOP Parts PFMEA																		
10	RM Procurement	MATERIAL FROM APPROVED SOURCE	MFC:-NIL SO:- Operation, Effected			Material from local market		Material from Approved source	Approved Source/ supplier list									
20	INCOMING INSPECTION	* THICKNESS OVER SIZE	MFC:-NIL SO:- Bending Dim. Effected CU:- Finished Issue EU:-NIL			1. Wrong PO / Incomplete Details in PO issued to supplier.		1. Verification by Sign. Authority of PO with Purchase indent and before issue to supplier.	1. MATERIAL CROSS VERIFY BY QA INSPECTOR BEFORE ENTERING THE PLANT AGAINST MTC & INVOICE.									
						2. Wrong TC Provided by Supplier.		2. Check at incoming stage	2. Mill TC should be verified by field inspector									
						3. No verification of TC by QA inspector.		3. Third party material test Plan to be adhered.	3. Verification of Mill TC material by MRN and third party test Report.									
						4. NON CALIBRATED INSTRUMENT USE FOR INSPECTION		4. Instrument calibration plan to be adhered.	4. Calibrated instrument should be used to check the material.									
		* THICKNESS UNDER SIZE	MFC:-NIL SO:- Bending Dim. Effected CU:- Finished Issue EU:-NIL			1. Wrong PO / Incomplete Details in PO issued to supplier.		1. Verification by Sign. Authority of PO with Purchase indent and before issue to supplier.	1. MATERIAL CROSS VERIFY BY QA INSPECTOR BEFORE ENTERING THE PLANT AGAINST MTC & INVOICE.									
						2. Wrong TC Provided by Supplier.		2. Check at incoming stage	2. Mill TC should be verified by field inspector									
						3. No verification of TC by QA inspector.		3. Third party material test Plan to be adhered.	3. Verification of Mill TC material by MRN and third party test Report.									
						4. NON CALIBRATED INSTRUMENT USE FOR INSPECTION		4. Instrument calibration plan to be adhered.	4. Calibrated instrument should be used to check the material.									
		* PLATE RUSTY	MFC:-NIL SO:- Poor surface Treatment CU:- Fail to sell EU:-Durability Effected			1. Material Stored in open condition.		1. Material Stored in covered area. Awareness about it to Quality inspector.	1. Incoming Inspection Report. Bottom steel process operator will ensure that Plate is free from RUST.									
						2. FIFO Not maintained		2. FIFO system applied	2. Color coding to be done on the steel plates. Green, Yellow.									
		* SHEET BEND	MFC:-NIL SO:- Bending Dim Effected CU:- Finished Issue EU:-Durability Effected			1. Plate may be fat / damage during unloading.		1. PROPER LOADING & UNLOADING BY HOIST (CRANE)	1. Incoming Inspection Report									
						2. Plate may not be stored properly		2. STORAGE LOCATION DEFINED FOR RAW MATERIAL	2. Before next process start operator should ensure that Plate in crt level / Inprocess inspection Report.									
		* MATERIAL GRADE WRONG	MFC:-NIL SO:- Bending Effected CU:- Rejection EU:-Durability Effected			1. Wrong PO / Incomplete Details in PO issued to supplier.		1. Verification by Sign. Authority of PO with Purchase indent and before issue to supplier.	1. Incoming Inspection Report									
						2. Wrong TC Provided by Supplier.		2. Verification of TC by Field inspector.										
						3. Chemical and Physical properties not matched with std		3. Chemical and Physical properties be checked & matched with std	3. Third party test to be done as per plan.									
30	Storage	* STORAGE ON WRONG LOCATION / MIXUP	MFC:-NIL SO:- Part may reject CU:- Fast Failure EU:-Field Failure			1. No proper place for storage of material by grade and thickness.		1. Storage area is identified as per grade and thickness area.	1. WI Displayed near Raw material Storage area and Incoming inspector Gives colour code after inspection.									
						2. No identification on sheets			2. Operator will use the material as per the color coding / FIFO system.									
40	Move for Next Process																	

ABBREVIATION- M/C:--MACHINE SO:-SECOND OPERATION CU:- CUSTOMER USE EU:- EXTERNAL USE

Prepared By : Mr.Amit Vishwakarma **Approved By- Mr.Prashant Raj Srivastava**

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17-Jul-25	A	Initial for revision no. "C"	Mr.Amit Vishwakarma	Mr. Prashant Raj Srivastava

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